



# **Numerical Optimization**

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### **Exercise sheet 4 for Numerical Optimization**

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# Chapter 1

## Solutions Exercise sheet 4

### 1.1 Exercise 1

#### 1.1.1 Part a)

**z Characterization of convexity by means of the first derivative** In this subsection we explain in detail what the statement of the exercise is trying to teach and then we prove both directions for parts (a) and (b). We work on an open and convex set  $C \subset \mathbb{R}^n$  and we assume that  $f : C \rightarrow \mathbb{R}$  is continuously differentiable.

$$f \text{ is convex on } C \iff f(x) - f(y) \geq \nabla f(y)^T(x - y) \text{ for all } x, y \in C. \quad (1.1)$$

The term  $\nabla f(y)^T(x - y)$  is exactly the value of the affine (linear + constant) function that represents the tangent hyperplane to  $f$  at the point  $y$ , evaluated at the point  $x$ . Thus (1.1) says: *the graph of a convex function always lies above all of its tangent hyperplanes*. Part (b) asks you to do the same but with strict inequalities in order to characterize *strict* convexity.

**Recall of basic notions.** First, recall the definition of a convex set:

$$C \subset \mathbb{R}^n \text{ is convex} \iff \forall x, y \in C, \forall \lambda \in [0, 1] : \lambda x + (1 - \lambda)y \in C. \quad (1.2)$$

Second, recall the definition of a convex function:

$$f : C \rightarrow \mathbb{R} \text{ is convex} \iff \forall x, y \in C, \forall \lambda \in [0, 1] : f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.3)$$

Third, for a differentiable function the gradient at a point  $y$ , denoted  $\nabla f(y)$ , is the vector of partial derivatives. For any direction  $x - y$ , the directional derivative of  $f$  at  $y$  in direction  $x - y$  is

$$Df(y)[x - y] = \nabla f(y)^T(x - y). \quad (1.4)$$

This is exactly the slope of the tangent in the direction from  $y$  to  $x$ .

We now prove the two parts.

**Convexity  $\iff$  first-order inequality** We must show that for a continuously differentiable function  $f : C \rightarrow \mathbb{R}$  on an open and convex set  $C$ ,

$$f \text{ is convex on } C \iff \forall x, y \in C : f(x) - f(y) \geq \nabla f(y)^T(x - y). \quad (1.5)$$

We do this in two directions.

(i) “ **$f$  convex**”  $\Rightarrow$  (1.5). Assume  $f$  is convex in the sense of (1.3). Let  $x, y \in C$  be arbitrary. Because  $C$  is convex, for every  $\lambda \in [0, 1]$  the point

$$y_\lambda := \lambda x + (1 - \lambda)y = y + \lambda(x - y) \quad (1.6)$$

belongs to  $C$ . By convexity of  $f$  we have

$$f(y_\lambda) = f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.7)$$

Subtract  $f(y)$  from both sides of (1.7) to obtain

$$f(y_\lambda) - f(y) \leq \lambda(f(x) - f(y)). \quad (1.8)$$

Now take  $\lambda > 0$  and divide (1.8) by  $\lambda$ :

$$\frac{f(y_\lambda) - f(y)}{\lambda} \leq f(x) - f(y). \quad (1.9)$$

Recall from (1.6) that  $y_\lambda = y + \lambda(x - y)$ . Since  $f$  is continuously differentiable, the limit of the difference quotient in (1.9) as  $\lambda \rightarrow 0^+$  is precisely the directional derivative of  $f$  at  $y$  in the direction  $x - y$ . Hence,

$$\lim_{\lambda \rightarrow 0^+} \frac{f(y + \lambda(x - y)) - f(y)}{\lambda} = \nabla f(y)^T(x - y). \quad (1.10)$$

Taking the limit  $\lambda \rightarrow 0^+$  on both sides of (1.9), and using (1.10), we get

$$\nabla f(y)^T(x - y) \leq f(x) - f(y), \quad (1.11)$$

which we can rewrite as

$$f(x) - f(y) \geq \nabla f(y)^T(x - y). \quad (1.12)$$

Since  $x, y \in C$  were arbitrary, (1.12) holds for all  $x, y \in C$ . This proves the direction “convex  $\Rightarrow$  (1.5)”.

(ii) (1.5)  $\Rightarrow$  “ **$f$  convex**”. Now assume that for all  $x, z \in C$  we have

$$f(x) - f(z) \geq \nabla f(z)^T(x - z). \quad (1.13)$$

We want to derive the usual convexity inequality (1.3). So let  $x, y \in C$  and  $\lambda \in [0, 1]$  be arbitrary. Define the point

$$z := \lambda x + (1 - \lambda)y. \quad (1.14)$$

Because  $C$  is convex, from (1.2) we know that  $z \in C$ .

Apply (1.13) once with the pair  $(x, z)$ :

$$f(x) - f(z) \geq \nabla f(z)^T(x - z). \quad (1.15)$$

Apply (1.13) again with the pair  $(y, z)$ :

$$f(y) - f(z) \geq \nabla f(z)^T(y - z). \quad (1.16)$$

Multiply (1.15) by  $\lambda$  and (1.16) by  $(1 - \lambda)$ , and then add the resulting inequalities. On the left-hand side we obtain

$$\lambda(f(x) - f(z)) + (1 - \lambda)(f(y) - f(z)) = \lambda f(x) + (1 - \lambda)f(y) - f(z). \quad (1.17)$$

On the right-hand side we get

$$\lambda \nabla f(z)^T(x - z) + (1 - \lambda) \nabla f(z)^T(y - z) = \nabla f(z)^T(\lambda(x - z) + (1 - \lambda)(y - z)). \quad (1.18)$$

But, using the definition of  $z$  in (1.14), we have

$$\lambda(x - z) + (1 - \lambda)(y - z) = \lambda x + (1 - \lambda)y - z = z - z = 0. \quad (1.19)$$

Therefore the right-hand side of (1.18) is

$$\nabla f(z)^T \cdot 0 = 0. \quad (1.20)$$

Combining (1.17) and (1.20) we obtain

$$\lambda f(x) + (1 - \lambda)f(y) - f(z) \geq 0, \quad (1.21)$$

which is equivalent to

$$f(z) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.22)$$

Finally, replacing  $z$  by its definition (1.14) we find

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y), \quad (1.23)$$

which is exactly the definition of convexity (1.3). Hence (1.5) implies that  $f$  is convex, and the proof of part (a) is complete.

### 1.1.2 Part b)

**Strict convexity  $\iff$  strict first-order inequality** We now prove the strict version. We want to show that

$$f \text{ is strictly convex on } C \iff \forall x, y \in C, x \neq y : f(x) - f(y) > \nabla f(y)^T(x - y). \quad (1.24)$$

Recall that strict convexity means

$$\forall x, y \in C, x \neq y, \forall \lambda \in (0, 1) : f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y). \quad (1.25)$$

Again we do both directions.

**(i) (1.24)  $\Rightarrow$  strict convexity.** Assume that for all  $x, y \in C$  with  $x \neq y$  we have (1.24). Let  $x, y \in C$  with  $x \neq y$  and let  $\lambda \in (0, 1)$ . Define

$$z := \lambda x + (1 - \lambda)y. \quad (1.26)$$

Then  $z \in C$ . Apply (1.24) with the pair  $(x, z)$ . Since  $x \neq z$  (because  $0 < \lambda < 1$ ), we obtain

$$f(x) - f(z) > \nabla f(z)^T(x - z). \quad (1.27)$$

Similarly, apply (1.24) with the pair  $(y, z)$ . Since  $y \neq z$ , we get

$$f(y) - f(z) > \nabla f(z)^T(y - z). \quad (1.28)$$

Multiply (1.27) by  $\lambda$  and (1.28) by  $(1 - \lambda)$  and add. On the left-hand side we get

$$\lambda(f(x) - f(z)) + (1 - \lambda)(f(y) - f(z)) = \lambda f(x) + (1 - \lambda)f(y) - f(z). \quad (1.29)$$

On the right-hand side we get, as before,

$$\lambda \nabla f(z)^T(x - z) + (1 - \lambda) \nabla f(z)^T(y - z) = \nabla f(z)^T(\lambda(x - z) + (1 - \lambda)(y - z)) = 0, \quad (1.30)$$

using exactly the same computation as in (1.19). But now both inequalities (1.27) and (1.28) were strict, so their positive combinations are still strict. Hence

$$\lambda f(x) + (1 - \lambda)f(y) - f(z) > 0. \quad (1.31)$$

Rearranging gives

$$f(z) < \lambda f(x) + (1 - \lambda)f(y), \quad (1.32)$$

and substituting (1.26) we obtain exactly (1.49). Thus (1.24) implies strict convexity.

**(ii) Strict convexity  $\Rightarrow$  (1.24).** Now assume that  $f$  is strictly convex in the sense of (1.49). From part (a) we *already know* that every convex differentiable function satisfies the non-strict inequality

$$f(x) - f(y) \geq \nabla f(y)^T(x - y) \quad \text{for all } x, y \in C. \quad (1.33)$$

Since strict convexity implies convexity, (1.33) is valid for our  $f$ . Thus we only need to show that for  $x \neq y$  equality in (1.33) cannot happen.

So let  $x, y \in C$  with  $x \neq y$ . Assume, towards a contradiction, that we have equality in (1.33), i.e.

$$f(x) - f(y) = \nabla f(y)^T(x - y). \quad (1.34)$$

Now consider the midpoint between  $x$  and  $y$ :

$$z := \frac{x + y}{2}. \quad (1.35)$$

Because  $C$  is convex,  $z \in C$ . Apply the non-strict inequality (1.33) to the pair  $(z, y)$ . We obtain

$$f(z) - f(y) \geq \nabla f(y)^T(z - y). \quad (1.36)$$

But

$$z - y = \frac{x + y}{2} - y = \frac{x - y}{2}, \quad (1.37)$$

so the right-hand side of (1.36) becomes

$$\nabla f(y)^T(z - y) = \nabla f(y)^T\left(\frac{x - y}{2}\right) = \frac{1}{2} \nabla f(y)^T(x - y). \quad (1.38)$$

Therefore (1.36) can be rewritten as

$$f(z) - f(y) \geq \frac{1}{2} \nabla f(y)^T(x - y). \quad (1.39)$$

Now use the assumed equality (1.34), namely

$$\nabla f(y)^T(x - y) = f(x) - f(y), \quad (1.40)$$

and substitute it into (1.39) to get

$$f(z) - f(y) \geq \frac{1}{2}(f(x) - f(y)). \quad (1.41)$$

This is equivalent to

$$f(z) \geq \frac{f(x) + f(y)}{2}. \quad (1.42)$$

But (1.42) contradicts strict convexity, because from (1.49) with  $\lambda = \frac{1}{2}$  we must have

$$f\left(\frac{x+y}{2}\right) < \frac{f(x) + f(y)}{2}. \quad (1.43)$$

Thus our assumption (1.34) that equality could hold is impossible. Therefore, for all  $x \neq y$ ,

$$f(x) - f(y) > \nabla f(y)^T(x - y), \quad (1.44)$$

which is exactly (1.24). This completes the proof of part (b).

We have shown:

$$(a) \ f \text{ convex} \iff \forall x, y \in C : f(x) - f(y) \geq \nabla f(y)^T(x - y), \quad (1.45)$$

and

$$(b) \ f \text{ strictly convex} \iff \forall x, y \in C, \ x \neq y : f(x) - f(y) > \nabla f(y)^T(x - y). \quad (1.46)$$

This is the first-order characterization of (strict) convexity.

## 1.2 Exercise 2

**(Strict / uniform) convexity of monomials** In this subsection we analyze the chain of notions *convexity*  $\Rightarrow$  *strict convexity*  $\Rightarrow$  *uniform convexity*, and we apply it to the monomials

$$p_n : \mathbb{R} \rightarrow \mathbb{R}, \quad p_n(x) = x^n, \quad n \in \mathbb{N}_0 = \{0, 1, 2, \dots\}. \quad (1.47)$$

The goal is to see precisely which exponents  $n$  give convexity, which give strict convexity, and which give uniform convexity.

**To consider:** Let  $C \subset \mathbb{R}^n$  be a convex set.

- A function  $f : C \rightarrow \mathbb{R}$  is called *convex* in  $C$  if for every  $x, y \in C$  and every  $\lambda \in [0, 1]$  it holds

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.48)$$

- The function  $f$  is called *strictly convex* in  $C$  if for every  $x, y \in C$  with  $x \neq y$  and every  $\lambda \in (0, 1)$  it holds

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y). \quad (1.49)$$

- The function  $f$  is called *uniformly convex* in  $C$  if there exists a constant  $\mu > 0$  such that for all  $x, y \in C$  and all  $\lambda \in [0, 1]$  it holds

$$f(\lambda x + (1 - \lambda)y) + \mu\lambda(1 - \lambda)\|x - y\|^2 \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.50)$$

Here  $\|\cdot\|$  denotes the Euclidean norm in  $\mathbb{R}^n$ .

Observe that (1.50) is strictly stronger than (1.48), because in (1.50) we are *adding* on the left-hand side the nonnegative term

$$\mu\lambda(1 - \lambda)\|x - y\|^2 \geq 0 \quad (1.51)$$

and despite that addition the inequality still holds.

**Illustrative difference (part (a)).** Convexity (1.48) says that the graph of  $f$  never lies above the chord joining the two points  $(x, f(x))$  and  $(y, f(y))$ . Strict convexity (1.49) says more: whenever  $x \neq y$  and the combination is *proper* (i.e.  $\lambda \in (0, 1)$ ), then the value of  $f$  at the middle point is *strictly* below the chord. Uniform convexity (1.50) says even more: for every two points there is a *quantified* gap between the curve and the chord, and the size of that gap is bounded from below by the quadratic quantity

$$\mu\lambda(1 - \lambda)\|x - y\|^2. \quad (1.52)$$

This means the bending of the function is not only present, but present with a *uniform* strength.

### 1.2.1 Part a)

**Implications** We now prove the chain

$$\text{uniform convexity} \Rightarrow \text{strict convexity} \Rightarrow \text{convexity}. \quad (1.53)$$

- (i) Assume  $f$  is uniformly convex, hence satisfies (1.50). Take arbitrary  $x, y \in C$  with  $x \neq y$  and arbitrary  $\lambda \in (0, 1)$ . Then

$$f(\lambda x + (1 - \lambda)y) + \mu\lambda(1 - \lambda)\|x - y\|^2 \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.54)$$

Because  $\mu > 0$ ,  $\lambda(1 - \lambda) > 0$  (since  $\lambda \in (0, 1)$ ), and  $\|x - y\|^2 > 0$  (since  $x \neq y$ ), we have

$$\mu\lambda(1 - \lambda)\|x - y\|^2 > 0. \quad (1.55)$$

Therefore, subtracting this strictly positive number from the right-hand side we get

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) - \mu\lambda(1 - \lambda)\|x - y\|^2 < \lambda f(x) + (1 - \lambda)f(y). \quad (1.56)$$

This is exactly the definition of strict convexity (1.49). Hence

$$\text{uniform convexity} \Rightarrow \text{strict convexity}. \quad (1.57)$$

(ii) If  $f$  is strictly convex, then for  $x = y$  or for  $\lambda = 0$  or  $\lambda = 1$  the inequality

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) \quad (1.58)$$

clearly holds with equality. For  $x \neq y$  and  $\lambda \in (0, 1)$ , strict convexity gives an even stronger inequality with a strict  $<$  sign. Therefore the weaker inequality (1.48) is always true, and so

$$\text{strict convexity} \Rightarrow \text{convexity}. \quad (1.59)$$

Thus the whole chain holds.

**Derivatives of monomials.** From now on we restrict to  $n = 1$  and to the real line  $\mathbb{R}$ . For the monomial

$$p_n(x) = x^n \quad (1.60)$$

we have, for  $n \geq 1$ , the first derivative

$$p'_n(x) = \frac{d}{dx}(x^n) = nx^{n-1}, \quad (1.61)$$

and for  $n \geq 2$  the second derivative

$$p''_n(x) = \frac{d}{dx}(nx^{n-1}) = n(n-1)x^{n-2}. \quad (1.62)$$

In one dimension, for  $C^2$  functions, the condition

$$f''(x) \geq 0 \quad \text{for all } x \in \mathbb{R} \quad (1.63)$$

is a standard sufficient (and on intervals, also necessary) condition for convexity. If in addition

$$f''(x) > 0 \quad \text{for all } x \in \mathbb{R}, \quad (1.64)$$

then  $f$  is strictly convex.

### 1.2.2 Part b)

**convexity of  $p_n$ .** We want to prove:

$$p_n(x) = x^n \text{ is convex on } \mathbb{R} \iff n = 0 \text{ or } n = 1 \text{ or } (n \geq 2 \text{ and } n \text{ even}). \quad (1.65)$$

- If  $n = 0$ , then

$$p_0(x) = 1 \quad \text{for all } x \in \mathbb{R}. \quad (1.66)$$

A constant function is convex, because for all  $x, y$  and all  $\lambda \in [0, 1]$  we have

$$p_0(\lambda x + (1 - \lambda)y) = 1 = \lambda \cdot 1 + (1 - \lambda) \cdot 1 = \lambda p_0(x) + (1 - \lambda)p_0(y). \quad (1.67)$$

- If  $n = 1$ , then

$$p_1(x) = x, \quad (1.68)$$

which is affine (linear plus constant), and every affine function is convex.



- Now assume  $n \geq 2$ . Then

$$p_n''(x) = n(n-1)x^{n-2}. \quad (1.69)$$

Since  $n \geq 2$ , the coefficient

$$n(n-1) > 0. \quad (1.70)$$

Hence the sign of  $p_n''(x)$  is completely determined by the sign of  $x^{n-2}$ .

- If  $n$  is even, write  $n = 2k$  with  $k \geq 1$ . Then

$$p_n''(x) = (2k)(2k-1)x^{2k-2}. \quad (1.71)$$

The exponent  $2k-2$  is even, because

$$2k-2 = 2(k-1). \quad (1.72)$$

Therefore, for every  $x \in \mathbb{R}$ ,

$$x^{2k-2} \geq 0. \quad (1.73)$$

Together with  $(2k)(2k-1) > 0$  we obtain

$$p_n''(x) = (2k)(2k-1)x^{2k-2} \geq 0 \quad \text{for all } x \in \mathbb{R}. \quad (1.74)$$

Hence  $p_n$  is convex on  $\mathbb{R}$  when  $n \geq 2$  is even.

- If  $n$  is odd and  $n \geq 3$ , write  $n = 2k+1$  with  $k \geq 1$ . Then

$$p_n''(x) = n(n-1)x^{n-2} = (2k+1)(2k)x^{2k-1}. \quad (1.75)$$

Now the exponent  $2k-1$  is odd. This implies

$$x^{2k-1} > 0 \quad \text{if } x > 0, \quad x^{2k-1} = 0 \quad \text{if } x = 0, \quad x^{2k-1} < 0 \quad \text{if } x < 0. \quad (1.76)$$

Since  $(2k+1)(2k) > 0$ , we get

$$p_n''(x) > 0 \text{ for } x > 0, \quad p_n''(x) = 0 \text{ for } x = 0, \quad p_n''(x) < 0 \text{ for } x < 0. \quad (1.77)$$

Thus the second derivative changes sign, so  $p_n$  cannot be convex on the whole  $\mathbb{R}$ .

Putting all cases together, we obtain exactly the statement of part (b).

### 1.2.3 Part b)

**strict convexity of  $p_n$ .** We show

$$p_n \text{ is strictly convex on } \mathbb{R} \iff n \geq 2 \text{ and } n \text{ is even.} \quad (1.78)$$

- If  $n = 0$ , then  $p_0$  is constant and therefore not strictly convex, because for strict convexity we would need a strict inequality.
- If  $n = 1$ , then  $p_1$  is affine and therefore not strictly convex.
- Let now  $n \geq 2$  be even,  $n = 2k$ . Then

$$p_n'(x) = \frac{d}{dx}(x^{2k}) = 2k x^{2k-1}. \quad (1.79)$$

Consider its derivative (which is  $p_n''$ ):

$$(p_n')'(x) = p_n''(x) = 2k(2k-1)x^{2k-2}. \quad (1.80)$$

Here  $2k(2k-1) > 0$  and  $2k-2$  is even, hence

$$x^{2k-2} \geq 0 \quad \text{for all } x \in \mathbb{R}, \quad (1.81)$$

and so

$$p_n''(x) \geq 0 \quad \text{for all } x \in \mathbb{R}, \quad (1.82)$$

with

$$p_n''(x) > 0 \quad \text{for all } x \neq 0. \quad (1.83)$$

The first derivative  $p_n'$  is therefore strictly increasing on  $\mathbb{R}$  (it increases on  $(-\infty, 0)$ , passes through 0 at  $x = 0$ , and keeps increasing on  $(0, \infty)$ ). A standard one-dimensional result states: if  $f'$  is strictly increasing on an interval, then  $f$  is strictly convex on that interval. Applying this to  $p_n$  we obtain that  $p_n$  is strictly convex.

- Conversely, if  $n$  is odd and  $n \geq 3$ , we already know  $p_n$  is not convex on  $\mathbb{R}$ , so it certainly cannot be strictly convex.

Hence part (c) is proved.

**Part (d): uniform convexity of  $p_n$ .** We prove

$$p_n \text{ is uniformly convex on } \mathbb{R} \iff n = 2. \quad (1.84)$$

A useful one-dimensional criterion is the following: if  $f \in C^2(\mathbb{R})$  and there exists a constant  $\mu > 0$  such that

$$f''(x) \geq 2\mu \quad \text{for all } x \in \mathbb{R}, \quad (1.85)$$

then  $f$  is uniformly convex on  $\mathbb{R}$  with that  $\mu$ . (In higher dimensions the condition is  $\nabla^2 f(x) \succeq 2\mu I$ .)

- For  $n = 2$ , we have

$$p_2(x) = x^2, \quad p_2''(x) = 2 \quad \text{for all } x \in \mathbb{R}. \quad (1.86)$$

Then (1.85) holds with  $\mu = 1$ , because

$$p_2''(x) = 2 \geq 2 \cdot 1 = 2\mu \quad \text{for all } x. \quad (1.87)$$

In fact, we can even verify the exact identity

$$(\lambda x + (1-\lambda)y)^2 + \lambda(1-\lambda)(x-y)^2 = \lambda x^2 + (1-\lambda)y^2, \quad (1.88)$$

which is precisely (1.50) with  $\mu = 1$ .

- For  $n = 0$ ,  $p_0''(x) = 0$  for all  $x$ , so there is no  $\mu > 0$  such that  $0 \geq 2\mu$ . Hence  $p_0$  is not uniformly convex.
- For  $n = 1$ ,  $p_1''(x) = 0$  for all  $x$ , so again not uniformly convex.
- For  $n \geq 3$  odd, we already know  $p_n$  is not even convex on  $\mathbb{R}$ , so it cannot be uniformly convex.
- For  $n \geq 4$  even, write  $n = 2k$  with  $k \geq 2$ . Then

$$p_n''(x) = 2k(2k-1)x^{2k-2}. \quad (1.89)$$

Evaluate at  $x = 0$ :

$$p_n''(0) = 2k(2k-1) \cdot 0^{2k-2} = 0. \quad (1.90)$$

Thus  $\inf_{x \in \mathbb{R}} p_n''(x) = 0$ . But uniform convexity would require the existence of  $\mu > 0$  such that  $p_n''(x) \geq 2\mu$  for all  $x$ , which is impossible because at  $x = 0$  we would need  $0 \geq 2\mu$ .

Therefore, the only exponent for which  $x^n$  is uniformly convex on all of  $\mathbb{R}$  is  $n = 2$ .

**From this exercise we could say:** For the monomials  $p_n(x) = x^n$  on  $\mathbb{R}$  we have:

$$\begin{aligned} \text{Convex on } \mathbb{R} &\iff n = 0, n = 1, \text{ or } n \geq 2 \text{ even,} \\ \text{Strictly convex on } \mathbb{R} &\iff n \geq 2 \text{ even,} \\ \text{Uniformly convex on } \mathbb{R} &\iff n = 2. \end{aligned} \tag{1.91}$$

This shows clearly the hierarchy

$$\text{uniformly convex} \Rightarrow \text{strictly convex} \Rightarrow \text{convex}, \tag{1.92}$$

and that for this simple family only the quadratic monomial  $x^2$  reaches the strongest level.

### 1.3 Exercise 3

#### Basic Definitions

1. **Convex set.** A set  $C \subset \mathbb{R}^n$  is convex if for all  $x, y \in C$  and every  $\lambda$  with  $0 \leq \lambda \leq 1$ , we have

$$\lambda x + (1 - \lambda)y \in C. \quad (1.93)$$

2. **Convex function.** A function  $f : C \rightarrow \mathbb{R}$  defined on a convex set  $C$  is convex if for all  $x, y \in C$  and all  $\lambda \in [0, 1]$ ,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (1.94)$$

Geometrically, the graph of  $f$  lies below the straight line joining the points  $(x, f(x))$  and  $(y, f(y))$ .

3. **Strictly convex function.**  $f$  is strictly convex if for all  $x \neq y$  and  $0 < \lambda < 1$ ,

$$f(\lambda x + (1 - \lambda)y) < \lambda f(x) + (1 - \lambda)f(y). \quad (1.95)$$

This means the function bends downward and is never flat between two different points.

4. **Local and global minima.**

- A point  $\bar{x} \in C$  is a **local minimum** if there exists a radius  $r > 0$  such that

$$f(\bar{x}) \leq f(x) \quad \text{for all } x \in C \text{ with } \|x - \bar{x}\| < r. \quad (1.96)$$

- A point  $\bar{x} \in C$  is a **global minimum** if

$$f(\bar{x}) \leq f(x) \quad \text{for all } x \in C. \quad (1.97)$$

5. **Stationary point (for differentiable  $f$ ).** If  $C$  is open and  $f$  is differentiable, a point  $\bar{x} \in C$  is called **stationary** if

$$\nabla f(\bar{x}) = 0. \quad (1.98)$$

#### 1.3.1 Part a)

**Every local minimum of  $f$  in  $C$  is a global minimum.** Let  $C \subset \mathbb{R}^n$  be convex and  $f : C \rightarrow \mathbb{R}$  convex. Assume  $\bar{x} \in C$  is a local minimum. We must show that

$$f(\bar{x}) \leq f(z) \quad \text{for all } z \in C. \quad (1.99)$$

Take any  $z \in C$ . If  $z = \bar{x}$  the inequality holds trivially, so assume  $z \neq \bar{x}$ .

Because  $C$  is convex, for every  $\lambda \in [0, 1]$  the point

$$x_\lambda := \lambda z + (1 - \lambda)\bar{x} \quad (1.100)$$

also belongs to  $C$ .

By convexity of  $f$  we have

$$f(x_\lambda) = f(\lambda z + (1 - \lambda)\bar{x}) \leq \lambda f(z) + (1 - \lambda)f(\bar{x}). \quad (1.101)$$

Since  $\bar{x}$  is a local minimum, there exists  $r > 0$  such that if  $\|x - \bar{x}\| < r$ , then

$$f(\bar{x}) \leq f(x). \quad (1.102)$$

Notice that when  $\lambda \rightarrow 0$ , we have  $x_\lambda \rightarrow \bar{x}$ . Hence, for sufficiently small  $\lambda$ , the condition  $\|x_\lambda - \bar{x}\| < r$  holds, so

$$f(\bar{x}) \leq f(x_\lambda). \quad (1.103)$$

Combining both inequalities gives

$$f(\bar{x}) \leq f(x_\lambda) \leq \lambda f(z) + (1 - \lambda)f(\bar{x}). \quad (1.104)$$

Subtracting  $(1 - \lambda)f(\bar{x})$  from both sides,

$$f(\bar{x}) - (1 - \lambda)f(\bar{x}) \leq \lambda f(z). \quad (1.105)$$

Simplifying the left-hand side,

$$\lambda f(\bar{x}) \leq \lambda f(z). \quad (1.106)$$

Since  $\lambda > 0$ , dividing by  $\lambda$  yields

$$f(\bar{x}) \leq f(z). \quad (1.107)$$

Because this holds for any  $z \in C$ ,  $\bar{x}$  is a global minimum.  $\square$

### 1.3.2 Part b)

**(If  $C \subset \mathbb{R}^n$  is open and  $f \in C^1(C)$ , every stationary point is a global minimum.** Assume  $C$  is open and convex,  $f : C \rightarrow \mathbb{R}$  is convex and differentiable, and  $\bar{x} \in C$  satisfies  $\nabla f(\bar{x}) = 0$ .

For differentiable convex functions, the following inequality holds for all  $x, y \in C$ :

$$f(y) \geq f(x) + \nabla f(x)^T(y - x). \quad (1.108)$$

(This means the graph of  $f$  always lies above its tangent hyperplanes.)

Setting  $x = \bar{x}$ , we obtain for all  $y \in C$ :

$$f(y) \geq f(\bar{x}) + \nabla f(\bar{x})^T(y - \bar{x}). \quad (1.109)$$

Since  $\nabla f(\bar{x}) = 0$ , this becomes

$$f(y) \geq f(\bar{x}) + 0^T(y - \bar{x}) = f(\bar{x}). \quad (1.110)$$

Thus,  $f(\bar{x}) \leq f(y)$  for all  $y \in C$ , which means  $\bar{x}$  is a global minimum.  $\square$

### 1.3.3 Part c)

**The set of solutions of (2) is convex (but may be empty).** The convex optimization problem is

$$\text{Find } \hat{x} \in C \text{ such that } f(\hat{x}) = \min_{x \in C} f(x), \quad (1.111)$$

where  $C$  is convex and  $f$  is convex.

Let the **set of optimal solutions** be

$$S := \{x \in C : f(x) = f^*\}, \quad (1.112)$$

where

$$f^* := \inf_{x \in C} f(x) = \min_{x \in C} f(x) \quad (1.113)$$

(assuming the minimum exists).

We must show that  $S$  is convex.

Take any  $x_1, x_2 \in S$ . Thus,

$$f(x_1) = f^* \quad \text{and} \quad f(x_2) = f^*. \quad (1.114)$$

Because  $C$  is convex, for all  $\lambda \in [0, 1]$  the point

$$x_\lambda = \lambda x_1 + (1 - \lambda)x_2 \quad (1.115)$$

belongs to  $C$ .

By convexity of  $f$ ,

$$f(x_\lambda) = f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2). \quad (1.116)$$

Substitute  $f(x_1) = f(x_2) = f^*$ :

$$f(x_\lambda) \leq \lambda f^* + (1 - \lambda)f^* = f^*. \quad (1.117)$$

By definition of  $f^*$  as the minimum value, we also have for any  $x \in C$ :

$$f(x) \geq f^*. \quad (1.118)$$

In particular, for  $x_\lambda$ :

$$f(x_\lambda) \geq f^*. \quad (1.119)$$

Combining both inequalities gives

$$f(x_\lambda) = f^*. \quad (1.120)$$

Therefore,  $x_\lambda \in S$ , and  $S$  is convex. It can, however, be empty if the minimum is not attained in  $C$  (for example,  $C = (0, 1)$  and  $f(x) = x$ ).  $\square$

### 1.3.4 Part d)

**If  $f$  is strictly convex in  $C$ , the problem has at most one solution.** Assume  $f : C \rightarrow \mathbb{R}$  is strictly convex on a convex set  $C$ . We must show that there cannot exist two distinct minimizers.

Suppose, to the contrary, that there are two different solutions  $x_1, x_2 \in C$ , with  $x_1 \neq x_2$ , such that

$$f(x_1) = f(x_2) = f^* = \min_{x \in C} f(x). \quad (1.121)$$

Because  $C$  is convex, for every  $\lambda \in (0, 1)$ ,

$$x_\lambda = \lambda x_1 + (1 - \lambda)x_2 \in C. \quad (1.122)$$

By strict convexity of  $f$ ,

$$f(x_\lambda) = f(\lambda x_1 + (1 - \lambda)x_2) < \lambda f(x_1) + (1 - \lambda)f(x_2). \quad (1.123)$$

Since  $f(x_1) = f(x_2) = f^*$ , we get

$$f(x_\lambda) < \lambda f^* + (1 - \lambda)f^* = f^*. \quad (1.124)$$

This contradicts the definition of  $f^*$  as the minimum value. Hence, two distinct optimal points cannot exist, and there is at most one solution.

**Keys (to remember) :**

- Convex domain + convex function  $\Rightarrow$  well-structured optimization problem.
- (a) Convexity removes “fake” local minima.
- (b) For differentiable convex functions,  $\nabla f = 0$  implies a global minimum.
- (c) The set of minimizers is convex.
- (d) If  $f$  is strictly convex, the minimizer is unique.

### 1.3.5 Part e - n)

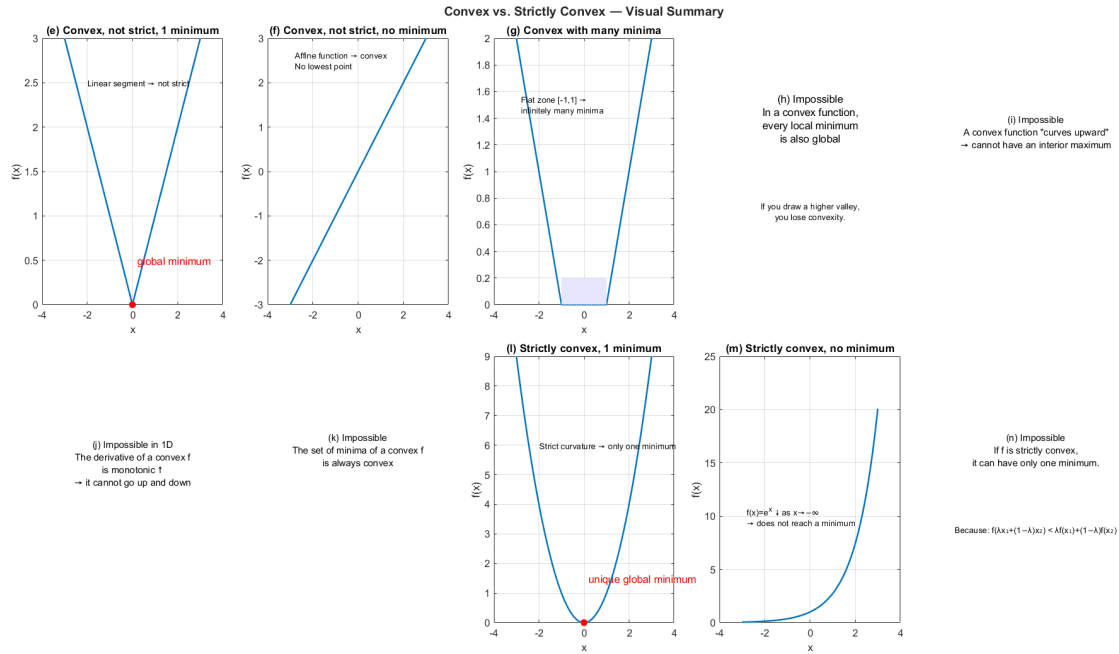


Figure 1.1: Functions of request

## 1.4 Exercise 4

### 1.4.1 Part a)

```

1
2
3 function[X] = hooke_jees(f, x0, h0, tol)
4
5 x = x0;
6 X = [x0];
7 h = h0;
8
9 while h>tol,
10     disp(x);
11
12     % perform exploration step
13     y=explore(f,x,h)
14
15     % if expl. step does not produce progress
16     if all(x==y)
17
18         % half step sizes and new expl. step
19         h=h/2;
20
21     % if expl. step does produce progress
22     else
23
24         % second exploration step
25         w=x+2*(y-x);
26         z=explore(f,w,h);
27
28         % if second expl. step better than first expl. step
29         if f(z)<f(y)
30             x=z;
31
32         % if not
33         else
34             x=y;
35         end
36
37         X = [X,x];
38
39     end
40
41 end
42
43 end
44
45 % exploration step in own function
46 function[x] = explore(f, x, h)
47
48     n = length(x);
49
50     % examine along coordinate axes of all n dimensions
51     for j=1:n
52         ej=zeros(n,1);
53         ej(j)=1;
54
55         % positive direction
56         t_pos=x+h(j)*ej;
57
58
59         %negative direction

```



```

60     t_neg=x-h(j)*ej;
61
62     %if decrease in positive direction
63     if f(t_pos)<f(x)
64         x=t_pos;
65
66     % if no decrease in positive direction, but decrease in negative direction
67     elseif f(t_neg)<f(x)
68         x=t_neg;
69     end
70 end
71 end);

```

hooke jeeves

### 1.4.2 Part b)

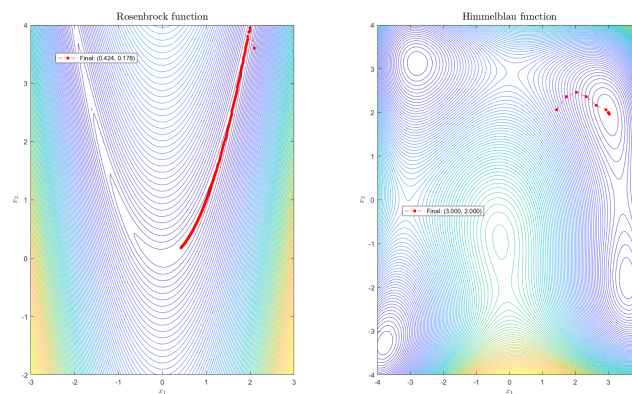


Figure 1.2: Hooke- Jeeves method on both Rosenbrocks and Himmelblaus function

The HookeJeeves method is a simple, derivative-free local search algorithm. It works efficiently for smooth functions with a single global minimum (like Rosenbrock), but when multiple local minima exist (like Himmelblau), it may converge to different solutions depending on the starting point. It is fast and easy to implement but sensitive to initialization and step size.

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## Exercise sheet 4 for Numerical Optimization

Dr. Michael Lehn

Tobias Born

Deadline: November 12, 2025

**Exercise 1 (Characterization of convexity on behalf of the first derivative, 5+4 points)**

Let  $C \subset \mathbb{R}^n$  be open and convex and  $f : C \rightarrow \mathbb{R}$  continuously differentiable. Prove that

- a)  $f$  is convex on  $C \Leftrightarrow$  for all  $x, y \in C$  it holds true that

$$f(x) - f(y) \geq \nabla f(y)^T (x - y) .$$

*Hint:* For  $\Rightarrow$  consider  $\lim_{\lambda \rightarrow 0} \frac{f(\lambda x + (1-\lambda)y) - f(y)}{\lambda}$  and for  $\Leftarrow$  consider  $z = \lambda x + (1-\lambda)y$  and show that  $\lambda(f(x) - f(z)) + (1-\lambda)(f(y) - f(z)) \geq 0$ .

- b)  $f$  is strictly convex on  $C \Leftrightarrow$  for all  $x, y \in C$  with  $x \neq y$  it holds true that

$$f(x) - f(y) > \nabla f(y)^T (x - y) .$$

*Hint:* For  $\Rightarrow$ : „ $\geq$ “ follows from a). Why cannot „ $>$ “ be proven considering  $\lim_{\lambda \rightarrow 0} \frac{f(\lambda x + (1-\lambda)y) - f(y)}{\lambda}$ ? Instead, assume „ $\geq$ “ and consider the point  $z = \frac{x+y}{2}$ .

**Exercise 2 ((Strict/uniform) convexity of monomials, 3+3+4+4 points)**

Let  $C \subset \mathbb{R}^n$  with  $C \neq \emptyset$  be convex. A function  $f : C \rightarrow \mathbb{R}$  is called *uniformly convex* in  $C$  if there is a constant  $\mu > 0$  such that for all  $x, y \in C$  and  $\lambda \in [0, 1]$  it holds true that

$$f(\lambda x + (1 - \lambda)y) + \mu\lambda(1 - \lambda)\|x - y\|_2^2 \leq \lambda f(x) + (1 - \lambda)f(y) . \quad (1)$$

- a) What is the illustrative difference between convexity, strict convexity and uniform convexity? Why do the implications *uniform convexity*  $\Rightarrow$  *strict convexity*  $\Rightarrow$  *convexity* hold true?

Consider  $p_n : \mathbb{R} \rightarrow \mathbb{R}$ ,  $p_n(x) = x^n$ , the monomial of order  $n \in \mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$ .

- b) Prove that  $p_n$  is convex on  $\mathbb{R}$  if and only if  $n = 0$ ,  $n = 1$  or  $n \geq 2$  and even.
- c) Prove that  $p_n$  is strictly convex on  $\mathbb{R}$  if and only if  $n \geq 2$  and even.
- d) Prove that  $p_n$  is uniformly convex on  $\mathbb{R}$  if and only if  $n = 2$ .

**Exercise 3 (Convex optimization problems, 3+3+3+3+1+1+1+1+1+1+1+1+1+1+1 points)**

A convex optimization problem is defined as

$$\text{find } \hat{x} \in \mathbb{R}^n \text{ such that } f(\hat{x}) = \min_{x \in C} f(x) \quad (2)$$

with convex domain  $C \subset \mathbb{R}^n$  and convex function  $f : C \rightarrow \mathbb{R}$ . We defined convexity of sets and functions in order to define convex optimization problems. These are fundamental for numerical optimization due to the following statements. Prove these statements.

- a) Every local minimum of  $f$  in  $C$  is a global minimum.
- b) If  $C \subset \mathbb{R}^n$  is open and  $f \in C^1(C)$ , every stationary point of  $f$  in  $C$  is a global minimum.
- c) The set of solutions of (2) is convex (but could be empty).
- d) If  $f$  is strictly convex in  $C$ , the convex optimization problem (2) has at most one solution.

In order to get a better intuition for these kind of problems, we now try to find simple, one-dimensional example problems.

- e) Draw a convex, but not strictly convex, function that has exactly one local minimum.
- f) Draw a convex, but not strictly convex, function that does not have a local minimum.
- g) Draw a convex function that has more than one local minimum.
- h) Try to draw a convex function having a local minimum that is not a global minimum and explain why this is not possible.
- i) Try to draw a convex function having a local maximum and explain why this is not possible.
- j) Try to draw a convex function having a saddle point and explain why this is not possible.
- k) Try to draw a convex function having a non-convex set of solutions of (2) and explain why this is not possible.
- l) Draw a strictly convex function that has exactly one local minimum.
- m) Draw a strictly convex function that does not have a local minimum.
- n) Try to draw a strictly convex function that has more than one local minimum and explain why this is not possible.

#### Exercise 4 (Direct search methods - Hooke-Jeeves method, 4+1 points)

Over the next weeks, we will be looking at unconstrained optimization, i.e., optimization problems without constraints. First we will have a look at *direct search methods* respectively *derivative-free methods*. These methods do not require any derivatives or regularities. They rely exclusively on evaluations of the objective function.

The idea is as follows. These methods can be used for functions that are not differentiable or whose derivatives are difficult to compute. However, one has to accept that in general there are no convergence statements and even if these methods converge, they usually perform worse than methods requiring more conditions.

Direct search methods are highly relevant for practical applications. Imagine objective functions that are complex black boxes, then it is difficult to approximate derivative values with acceptable accuracy and effort.

The first direct search method is the *Hooke-Jeeves method*. For  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ , the idea is to try to find its minimizer by successively searching along all coordinate axes. In each iteration step, an *exploration step* is performed to find a direction in which the function decreases comparatively rapidly. In the subsequent *progression step*, the next iterate is determined based on the direction from the exploration step.

Exploration step: Consider a current iterate  $x^{(k)} \in \mathbb{R}^n$  and a vector of step sizes  $h \in \mathbb{R}_+^n$ , where  $h_j, j = 1, \dots, n$ , is the step size along the coordinate axis of the  $j$ -th dimension. Then, one after the other, the directions of the coordinate axes of all  $n$  dimensions are examined. For the  $j$ -th dimension it is checked whether

$$f(x^{(k)} + h_j e_j) < f(x^{(k)}) ,$$

where  $e_j$  is the  $j$ -th unit vector of  $\mathbb{R}^n$ . If so, then  $h_j e_j$  is added to the iterate. If not, then  $x^{(k)} - h_j e_j$  is examined and if

$$f(x^{(k)} - h_j e_j) < f(x^{(k)}) ,$$

then  $-h_j e_j$  is added to the iterate. After having tested all  $n$  dimensions, we obtain

$$y^{(k)} \quad \text{with} \quad y_j^{(k)} \leftarrow x_j^{(k)} + \begin{cases} h_j e_j & \text{first check successful} \\ -h_j e_j & \text{first check not successful, second check successful} \\ 0 & \text{both checks not successful} \end{cases}$$

and it holds true that

$$f(y^{(k)}) \leq f(x^{(k)}) .$$

If  $y^{(k)} = x^{(k)}$ , the exploration step did not produce progress, the step size is set to half and a new exploration step is carried out. If  $y^{(k)} \neq x^{(k)}$ , a direction of decrease is found and the subsequent progression step is carried out.

Progression step: The direction of decrease is  $y^{(k)} - x^{(k)}$ , so you could actually search along the straight line

$$x^{(k)} + t(y^{(k)} - x^{(k)}) , \quad t > 0$$

for the lowest function value and select this point as the next iterate  $x^{(k+1)}$ . But how can one do this line search without having any derivatives of the objective function available? For this reason, the line search is usually replaced by setting  $t = 2$ , which results in

$$w^{(k)} \leftarrow x^{(k)} + 2(y^{(k)} - x^{(k)}) = 2y^{(k)} - x^{(k)} ,$$

and then performing a further exploration step starting from  $w^{(k)}$ . We denote the result of this second exploration step as  $z^{(k)}$ . If the second exploration step yields an improvement, so  $f(z^{(k)}) < f(y^{(k)})$ , the new iterate is set to  $z^{(k)}$ . If not, it is set to  $y^{(k)}$ .

The complete algorithm can be found in the lecture notes.

- a) We want to implement the Hooke-Jeeves method. To do this, fill in the gaps in the MATLAB programme `hooke_jeeves.m`.
- b) The MATLAB programme `test_hooke_jeeves.m` tests your implementation of the Hooke-Jeeves method on both Rosenbrock's and Himmelblau's function. Execute it and reflect on the results.